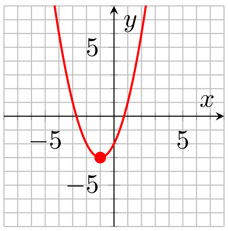
MTH 137(B) Dr. James Ashe
Quiz 2.1-2.2 Monday 4$^{\text{th}}$ March, 2013 **Name: Solutions**

Determine the domain and range of each relation, and state whether the relation is a function.

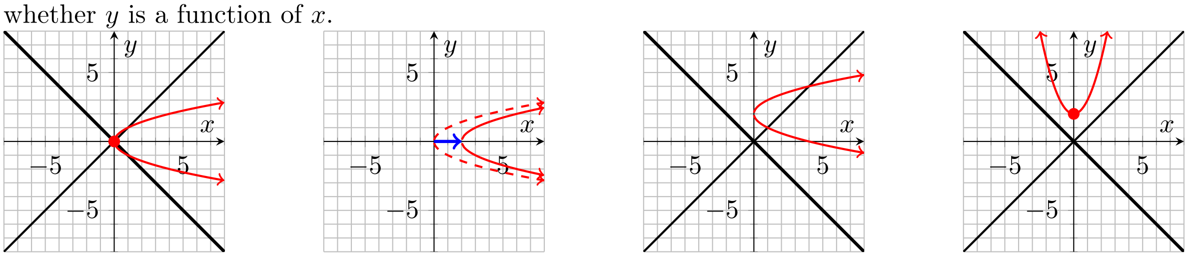
1. $f = \{(10, 1), (27, -1), (37, 1), (10, 2), (47, 2)\}$
No, it is not a function. The point 10 is mapped to 2 and 1.
Domain: $\{10, 27, 37, 10, 47\}$
Range: $\{1, -1, 2\}$

2. The curve shown in the graph below.



Yes, by the vertical line test it is a function.
Domain: $(-\infty, \infty)$
Range: $[-3, \infty)$

3. $x = y^2 + 2$. Choose the correct graph of the equation, determine the domain and range, and determine whether y is a function of x.



No, by the vertical line test it is not a function.
Domain: $[2, \infty)$
Range: $(-\infty, \infty)$

Let $g(x) = -x^2 + 5x + 3$.

4. Find $g(-2)$.
$g(-2) = -(-2)^2 + 5(-2) + 3 = -4 - 10 + 3 = -11$

Use the difference quotient, $\dfrac{f(x+h) - f(x)}{h}$ *for* $h \neq 0$*, to complete and simplify the following problem.*

5. $f(x) = x^2 + 1$.

$$\frac{\left[(x+h)^2 + 1\right] - \left[x^2 + 1\right]}{h} = \frac{x^2 + 2xh + h^2 + 1 - x^2 - 1}{h} = \frac{2xh + h^2}{h} = 2x + h$$